02 - Dataset Understanding

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Dataset source and dimensions

The repository uses the IBM Telco Customer Churn dataset distributed as WA_Fn-UseC_-Telco-Customer-Churn.csv. The EDA notebook reports 7,043 rows and 21 columns. After converting TotalCharges and dropping rows that become missing, the model-building data contains 7,032 observations.

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2. Feature groups

Group	Columns / examples	Meaning
Customer demographics	gender, SeniorCitizen, Partner, Dependents	Basic customer profile
Account / subscription	tenure, Contract, PaperlessBilling, PaymentMethod	Customer relationship and billing arrangements
Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Services subscribed to
Billing	MonthlyCharges, TotalCharges	Current monthly charge and cumulative charges
Identifier / target	customerID, Churn	Unique identifier and outcome label

3. Target distribution

The EDA notebook reports an approximate 73:27 distribution: about 73% non-churn and 27% churn. Exact counts should be recalculated from the cleaned dataset rather than inferred from the rounded percentages.

- Class 0: customer did not churn.
- Class 1: customer churned.
- The target imbalance makes accuracy alone an insufficient evaluation measure.

4. Important data-type observation

TotalCharges is initially read as an object/string column even though it represents a numeric amount. The notebook converts it to numeric, coercing unparseable blank entries to missing values. SeniorCitizen is numerically encoded as 0/1 but should be interpreted as a categorical indicator in descriptive analysis.

5. Interview-ready explanation

> *We used the IBM Telco Customer Churn dataset, containing 7,043 customer observations and 21 columns. It included demographic information, services subscribed, contract details, tenure, and billing attributes. Churn was the binary target. The target distribution was imbalanced, with roughly 73% non-churn and 27% churn observations, which influenced our evaluation strategy.*

6. Questions to prepare

Q. Why remove customerID?

It is an identifier, not a meaningful customer behaviour attribute. Including it can let a model memorize arbitrary identifiers or exploit accidental patterns.

Q. Why is TotalCharges initially an object?

The CSV contains blank or non-numeric values. Pandas therefore reads the column as text; it must be parsed before numeric analysis or modelling.

Q. Why is class imbalance relevant?

A model can achieve seemingly good accuracy by predicting the majority class while missing many churners. Recall, precision, F1, and precision-recall analysis are important.